

Dataset: <https://www.kaggle.com/datasets/shengkunwang/housets-dataset>

Ideas:

1. How did home value recovery after COVID-19 differ across the 30 major U.S. metro areas?
2. Do ZIP codes with more amenities (parks, schools, restaurants) consistently show higher or faster-growing home values?

One line code to read data: `awk -F, '{print $1,$2}' HouseTS.csv`

Descriptions of the variables:

1. Housing-market indicators (monthly):
Including variables about dynamic pricing and market activities such as sale and listing prices, inventory levels, new listings, etc.
2. Points of Interest (monthly):
Including variables about amenity based features such as restaurants, schools, parks, supermarkets etc.
3. Socioeconomic indicators (annual, temporally aligned to monthly):
Including variables about income, employment, population, education, and demographic characteristics.

Statistical methods:

1. Covid recovery
 - time-series decomposition: separate ZHVI in trend, seasonality, and noise to drag out the Covid dip and recovery
 - percent change calculation: measure each metro's drop (eg. Feb to May 2020) and recovery speed (months to return pre-covid ZHVI)
 - ANOVA: formally test whether recovery rates are statistically different across metros
 - K-means clustering: group the 30 metros into clusters like "fast/slow recovery"
2. Amenities vs. home value
 - PCA: reduce amenity types to a few key dimensions first before regression
 - correlation analysis: measure which amenity types (parks, schools, restaurants) have the strongest relationship with ZHVI levels
 - multiple linear regression: model ZHVI growth as a function of amenity counts, controlling for other variables like income and location

Computational steps:

The computational steps include acquiring&reading data, splitting data, constructing new features, parallel computation, summarizing & visualising results.

